

Active Inference BookStream #001.07 ~ "Governing Continuous Transform

BookStream | 51:53

Hello and welcome. It's January 24th, 2023. We're in ACT-IMP bookstream number 001.07 on Governing Continuous Transformation, off to you, Blue.

Great. So we are the Active Inference Institute, a participatory online institute that is communicating, learning, and practicing applied to Active Inference. You can reach us at all of these social media links. This is a recorded and archived live stream, so please provide us with feedback so that we can improve upon our work. All backgrounds and perspectives are welcome here, and we will be following good video etiquette for live streams. Again, if you'd like to participate, activeinference.org is our main website, and there's a link to get involved and join in many of the ongoing collaborative projects or live streams or both. This is, as Daniel said, bookstream 1.07, so it's our seventh contextualization video on this book, Governing Continuous Transformation, by Bijan Kesri. So we will just jump right into it. I'm Blue. I'm an independent researcher in New Mexico, and I will pass it off to Daniel. All right. I'm Daniel. I'm a researcher in California, and just happy to see this series continue. Tyler? And I'm Tyler. I'm a protocol designer and now researcher in California. Great. So we are in part two, free energy governance of this textbook, and the name of this chapter is Cognition from Superposition to Reality. And so I think that that kind of hints to some quantum stuff that might be elaborated more towards the end of this chapter and in the next chapter. And just a few keywords for this chapter, but still quite a bit of content here. So Tyler, do you want to read the start of the abstract? Do you mind? Sounds good. Can you guys hear me okay? Great. All right. Attention-based view and an activist approach address FEG's cognition dimension. To this day, strategy management research is not entirely detain from its very path dependency, but putting perception ahead of the action, i.e., putting environment ahead of strategy. While ABV's structure cognition nexus is FEG's critical building block, ABV erroneously treats action as a consequence of perception. Once we embrace attention as action generated, we start understanding that clever analysis and form of belief updating follows action generated stimuli, not the other way around. ABV is trapped in the old school limitations opposed by bounded rationality. In the discontinuous and distributed quantum world, agents' bounded rationality is meaningless, as enacted inference empowers the world to deliver a supervisory signal. Effectively, agents offload the burden of cognition to the environment. Bounded rationality is replaced by extended cognition. The boundedness of human cognition is effectively neutralized. Blue, do you want to take the next one? Sure. There are at least as many reality models as there are economic actors. Every organization chooses akin to holding, acting upon, a light torch in the darkness, which environmental significance to attend to. Whatever the torch highlights is equivalent to breaking a quantum wave of probabilities, moving from superposition to reality, intermediated by the Markov blanket. Environmental enactment is essentially a purpose-directed, interactive, inferential, and social learning sense-making process. All right, so there are two sections to this chapter. The first is attention-based view,

and the second is inactivist approach. So, Tyler, do you want to take this slide?

Sure. All right. So attention-based view. So we actually touched on this a couple of streams ago in the context of when we introduced this was that Bajan was going through different models of thinking about firm cognition, and they're all limited in their own sense, including attention-based view. And we'll go deeper into why attention-based view is limited, but also has important ways of thinking about how firm cognition works. And so the general idea of it is that attention is a limited resource, and that permeates through all levels of the hierarchy of an organization. So at the individual level, you have limited attention that bounds the level, that bounds your cognition of what you're able to understand and perceive and decisions you're able to make. And the boundedness of your individual cognition bubbles up into a social context, which bubbles the kind of decisions a group of people can make. And then that finally bubbles up into your organizational cognition, which manifests through different rules, positions, players, and resources. So this is just to say, like, hey, we only have so many things that we can spend our time and energy on, and that manifests at larger levels in the kinds of rules that we make within an organization. Blue, Danielle, anything you would add here? So I can read this claim. While attention-based views structure cognition nexus is a critical building block of free

energy governance, fundamentally, attention-based view suffers from severe shortcomings that weaken its relevance to

practice. By highlighting some of attention-based views most fundamental weaknesses, we should further cement the case for free energy

governance as a meta theory connecting in circular causality structure, cognition, and capabilities.

Once we understand how the Bayesian brain is building a model of its world, we start seeing how anachronistic and full of friction

our deep-rooted conceptions of organizational attention are.

So, do you guys have anything to add here? Okay, cool. All right. So, bounded rationality. So this was a concept that

Bijan brought up here in this chapter, and it was introduced by Herbert Simon, the term, in, like, 1957, and I pulled this from the

Stanford Encyclopedia of Philosophy. So his introduction of this term was a shorthand for his position against neoclassical economics, neoclassical economics, and his call to replace the perfect rationality assumptions of homo economicus with the conception of rationality tailored to cognitively limited agents.

And so, like, this little figure here shows, like, the cognitive limitations and imperfect information and time constraints that we're faced with when we have to make a decision. So we don't, we always, like, are limited by what we can hold in our brain and, like, the view of the information that we have. And, like, we were talking about, like, complexity, like computing pi, like, we could get a more accurate, you know, numerical digit of pi, but we're limited, like, by time. Like, if we go on and on and on, we could get more and more accurate, but, like, why? Like, is 3.14 good enough? Because I could pull it up in, like, a split second from memory. And so all of this, like, configures our bounded rationality. And we use a process that Herbert Simon introduced as satisficing to make a suboptimal decision or,

like, our best possible decision with the bounded rationality that we're faced with.

And a quote from the Herbert Simon says, broadly stated, the task is to replace the global rationality of economic man with the kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms, including man, in the kind of environments in which such organisms exist. Tyler, do you want to read where this bounded rationality concept is introduced in the chapter?

Sure. Yeah. So I think that the context here is that he's trying to connect

ABV to bounded rationality by saying, all right, well, you have limited attention. How does a firm decide where its attention is spent? Right? Like, how do you make that decision collectively? So he says, in ABV, top leadership's attention to the right issues and solutions considered the source, the firm's competitive advantage, essentially, is content rather than process centric. And the firm's performance bottleneck remains upper echelon's attention, i.e. bounded rationality. The critical question then is how, in the first place, context is selected to determine attention or how bottom-up market signals connect with top-down, upper echelon expectations and predictions to determine attention. And so what he's saying is like ABV doesn't really explain how that attention gets directed in the first place. ABV is trapped in the old school limitations imposed by bounded rationality and discontinuous and distributed quantum world agents bounded rationality is meaningless as perceptual content is given in our predictions. This is precisely what allows the world to deliver the supervisory signal. Effectively, the firm is offloading the brilliant condition to its environment. So bounded rationality is extended, is replaced by extended cognition. The boundedness of human cognition is effectively neutralized. Blue, Danielle, anything to add there?

Yeah, we'll go to the next slide where Blue could explain more about extended cognition.

Cool. So extended cognition, I took this from this article by Mark Sprivak. Extended cognition takes the idea that your mind is on your smartphone, literally. So it says that human cognitive states and processes sometimes spill outside of our heads into objects in our environment. And this is something that we've looked at and talked about before in many ways, like even with Avelle, I'm not sure if you guys talked about it yesterday on the stream, but in the thinking like a state livestream that we did, we talked quite a bit about extended cognition and cumulative cultural evolution and how like the, a lot of our cognition is like out there embedded in our environment so that we don't have to think about it. Like when I grew up, there was no internet. So we would have to like, you know, I mean, Colton, like my, my son in the car the other day asked me some question and I was like, you're going to have to ask Alexa or Google. And, and he was

like, will you ask your phone for me? And, um, it was like some fact question, like how many bones are in the body or

something that I couldn't remember off the top of my head. And, um, I said, well, wait till we get to school and then I'll ask my phone.

Cause my phone was like, whatever, playing music or doing, and I'm driving. Right. Um, and he was like, that was so long. Like, like you can't wait 15 minutes to get to school to like answer the question.

And I was like, man, when I was a kid, if I wanted to know the answer to this question, like I would have to go to the library, look in the card catalog, find a book like an encyclopedia or something. Oh, it's a Guinness book, a world record. That's what it was. It was like the world's smallest pancake or something like that.

Um, and so like, we would have to find the book and then look it up in the book and it would take days. Like if we could even remember the question for that long. And he was like, Oh my God, why would you live? So like this kind of extended cognition, like we could just know everything with a click of a few buttons is amazing. Um, do you guys have anything to add here at extended cognition? Sorry, I rambled.

[illegible]

So what Bajan says is while ABV views organizations as systems of structurally distributed attention, where structure channel attention, ABV treats the process of channeling as a black box.

So it's like a framework for thinking about how cognition is limited in organizational context, but it doesn't really tell you like what's going on.

For that reason, it doesn't end up being super practical, at least in Bajan's telling.

So he goes on to say neither UEV nor ABV, both fundamentally top-down oriented frameworks, have given due consideration to the dynamic interplay between top management team, TMT, and the board of directors.

More specifically, there's no transparent structural link between the upper echelon level and field intelligence, for example.

The very process that governs the two layers of interaction determining affirms attention and actions remains opaque.

And so I think broadly what Bajan's saying is that, especially like the inactivist approach of thinking that how your actions in the environment determine your perception, that's missing in ABV, and that's a critical aspect of understanding how cognition happens in an organization, but specifically the interplay between the board and the management team, which in previous streams, he spent a lot of time describing why that's a really critical and important relationship for firm cognition.

Blue, Danielle, anything you'd add here?

So I would just add that like, or maybe emphasize the point that we talk about, a lot about top management tier and the board and that interaction, and then like the interactions between different levels of the hierarchy, like from the top all the way to the bottom.

And I think that we emphasize the importance of like the feeding forward and feeding back and like the interaction between the layers and how information is transmitted between them, but there's no real concrete examples of how to best do that.

And I think that this is like, for me,
like not just in a firm, but also like in the body,
like how is that information like transmitted?

Like how does like my big toe prioritize,
like what goes to my brain?

Like obviously if I kick something really hard
and hurt my toe, or like if I get like a hangnail
or like ingrown toenail or something, it really hurts.

But like, otherwise my toe,
like I don't even know it's ever talking to my brain.

So it has to like provide salience somehow
and how this is done.

And I don't know, that's so critical for me.

I think that that's like really at the heart of all
of the active inference hierarchical structuring stuff.

All right.

Attentional coherence.

So here, Bijan quotes Ocasio,
and this is like a 2011 paper, I guess, or maybe book,
but he puts a lot of emphasis on Ocasio
as like one of like the people that like his work
really rests strongly upon.

So I encourage you to read the chapter
if you want to learn more or go straight to Ocasio itself.

So he says, the interaction of top down
and bottom up attentional processes is also implicit
in the Wick and Sutcliffe concept of the quality
of organizational attention and in the re-rope analysis
of attentional triangulation, combining attentional stability,
top down, attentional vividness, bottom up,
and attentional coherence, combining both.

And he goes on to say that attention-based view cannot explain
attentional coherence.

And so getting at what I just was saying about like the interaction
between the top down and bottom up and like how that works,

can be explained in attention-based view.

Maybe it can be explained in free energy governance,
or we can maybe get some guidance on how to best do that.
Hopefully in the coming chapters, we'll learn more.

Do you guys have any comments here?

Okay.

Do you want to read these claims for me, Tyler?

Sure.

So ABV erroneously treats action as a consequence of attention
and perception.

According to PPF, prediction, processing framework,
action is determined in circular causality to stimuli
that dictate attention and sense-making.

While ABV is laying a critical foundation
in looking cognition and structure, the question arises
whether ABV's attention will need to be redefined
as something more granular and practical.

So attention not only is too diffuse as a cognitive concept,
but also is hard to meaningfully operationalize
in the context of continuous strategic renewal.

So exactly what Blue was saying on the previous slide.

It doesn't, ABV explains like, okay, I understand why
attention being limited shapes a firm's cognition,
but like, what do I even do with that information?

Like, how is that gonna be impactful?

How can I model that?

And Bajan is planning like, well, it's not that useful right now,
but he's teasing that maybe it could be with active inference
or free energy governance.

Yeah. Like, what is that something more granular and practical?

Like, how do we focus attention?

Like, how do we like break attention up into like,
sub components that are more granular and practical?

Like, yeah.

Like what is in our field of view or what should be
in our field of view?

Like what we're looking at, what we're prioritizing
and do those things line up?

I don't know.

I could speculate all day, but it's hard to know really what that more granular and practical thing might be.

One other quick note on that is there's the question of coherence and adaptivity.

For example, that student answering their cell phone in the classroom on the previous cartoon, that may have certain dimensions of adaptivity or fulfill certain preferences of one agent, but not another.

And then there's this question of action orientation.

And so if we're always in the business of asking about the environment, doing the market research, not thinking about the strategy, taking this sort of perception first mindset, there still is the question of attentional coherence.

Are we paying attention and sense-making, putting all the pieces together from what we're perceiving?

And also there is this inactive shift, which is incomplete, but it's part of the puzzle, which is we want to be having attention, regimes of attention on our action.

And that brings together the embodiments and the inactive view with the attentional view, which as brought up, doesn't really describe what attention is and still can be mobilized in a passive setting.

But together we have this active attention and that flowers in active inference where attention is internal action.

So is that like pay attention to what you're paying attention to?

Is that like a nested little hint there?

Well, in live stream 28, we saw a three layer model of attention and there is no single one way to model any cognitive phenomena.

It's going to be up to people in different settings to understand what treatments of attention are insightful or useful or applicable.

I don't know.

I think that that like paying attention to what you're paying attention to kind of gets

like a lot of attention in like the business context

or like even in like prioritizing like your to do lists, right?

Like, are you just like fighting fires all the time

or are you like making progress to do like things going forward,
making progress on your own goals?

So I think like, I don't know, that might be might be a critical
like governing variable, right?

All right.

So I'm going to read this claim in that regard.

Predictive processing framework in in the form of inactive inference
is potentially a critical complement to and possibly substitute
for the attention based view.

Predictive processing framework provides the concepts and logics
that give new meaning to the linkage of structure and cognition.

According to predictive processing framework, prediction models
and predictions self optimize as a matter of pursuing surprise
minimization by way of free energy minimization.

It is strictly a priori, not an attention challenge,
but a purpose directed prediction challenge.

The evolution of self driving vehicles provides an exemplary illustration of this point.
Once the challenge was conceived as a prediction rather than attention challenge,
the road to perfection is simply a matter of continuous
and purpose directed prediction error minimization.

And I don't know, like, what is really meant by this?

Like, like, the challenge of self driving was conceived as a prediction,
like once people predicted that it would happen.

I'm not really sure, like what Bajan means here.

I could maybe take a stab at that one.

Because I think what he's saying is that, like, for a self driving car,
that's the kind of thing that if you were being using purely linear thinking,
only using the things you know already about the world,
like you would never, a self driving car wouldn't be like an obvious next step.

Like you have to take this leap of faith and action oriented leap of faith
and just saying, no, we're going to build a self driving car.

And then once you've decided that that's your path, you're going to go down.

Now it's just like optimization on that path.

And so I think this, he was like illustrating the inactivist approach of like,
hey, a self driving vehicle, there's nothing like obvious or self evident

about that needing to happen.

But once you decide it's going to happen, then it's a prediction problem.

Agreed.

There's different readings.

One way, also not knowing the self driving car game.

If we were in this passive perception first world,

then there would be the main issue to resolve would be understand from noisy sensor readings,

where is the location of a person and producer uncertainty about where the person is

or what that object is.

However, once we see it in terms of this continuous purpose directed prediction error minimization,

we're going to center the role of action, gas brake and turning.

And a few other affordances that cars have.

So instead of going for all the decimal points on whether that's a person or a mailbox or a cone,

what does it look like to take this unfolding action oriented view,

where you're aligning with your preferences, for example, not to cause harm to people,

and then lesser priority not to cause harm to objects.

Maybe steering away from that object, you can have a good policy posterior on.

It's a good and likely action without having all the decimals of precision about the environment.

So once we understand how essential action is to the self-driving car,

it reframes what we have to do with the sensor data,

and it centralizes the relevance of the strategy.

Yeah, I do remember something about like some change in self-driving like in the sensor system.

So maybe that like references the like attention versus prediction problem,

or maybe that's like being referred to internally in there.

I'm not sure.

Okay, so that sums up the attention-based view section,

and we're moving in now to the inactivist approach.

So this is at the start of the chapter.

Bajan quotes Thompson 2007 and says,

the organism's environment is the sense it makes of the world.

The environment is a place of significance and valence

as a result of the global action of the organism.

So again, taking this like action-centric perspective.

And he quotes Ramstad et al. 2020 by saying active inference is an active inference.

And he says,

Indeed, the essential question is how the firm brings forth significance and valence

that does not preexist, but is the result of action-centric

and socially constructed engagement with the outer world,

equivalent to internally assigning probabilities to the external,
hidden unknown states in the form of a Markov blanket.

This revealing of significance and valence then lays the foundation
for generating shared meaning and shared expectations
in the form of unique interpretation models, belief systems,
underlying firm performance, and survival.

So, yeah.

So, we can go on to see what else he says.

So, he makes the claim,

Dafton Week 1984 differentiate between environment-driven
and interpretation-driven approaches to strategy making.

And in this big quote, they say,

In environment-driven frames, top managers experience environments as concrete,
hard, measurable, and determinant.

On the other hand, top managers of firms confronting more unstable
or unanalyzable environments are thought to construct,
coerce, or enact a reasonable interpretation
that makes previous action sensible.

Thus, they attempt to construct their environments through their strategies
rather than developing strategies in response to environments,
resulting in strategy, environment beliefs, or proactive logics.

And he also says,

Despite WIC's strong influence on strategic management research to this day,
the environment-driven approach remains dominant.

So, I think that this is like one of the central themes that's emerging in the book
is that like the environment is somewhere out there.

It's some very real thing.

It's taken to be like it is this way.

It's always going to be this way.

And it's like and kind of something that Daniel and I have talked about at length,
like the niche as responding to the organism,
the niche response to the organism just as much as the organism responds to the niche.
And whether or not the niche includes other organisms in the environment or not
is kind of irrelevant when we have things like books and smartphones.

So, this environment is changing,

and it's changing maybe so fast and so much that this environment-driven framework
is perhaps not effective.

So, unstable or unanalyzable environments is maybe like this discontinuous environment

that Bijan speaks of because the environment's changing fast,
like maybe faster than it's ever changed before
because of like our global connectivity and many other reasons.
So, now maybe all environments are unstable or unanalyzable
or more unpredictable than they've been before
so that this constant update or attempt to attempting to construct the niche
rather than viewing it as something stable
is maybe more important for firm persistence.

Do you guys have any comments here?

Yeah, I mean, I just think on the second half of that,
of like very unstable and unanalyzable environments,
that really resonates in the startup context
where a lot of times startup founders will have no pretense of knowing
anything that's going on in the environment.

Like, they're very clear about their ignorance of the world.
Instead, they run small experiments to kind of gauge what's going on
in the external world
and then kind of construct their worldview from there
rather than saying, hey, we have this great business idea.

Let's scale this thing.

It's more like, hey, let's discover who our users are,
which I think really corresponds well with that second half of the quarter.

Also, with our classical temperature example,
let's just say that we were in a house and there was a heating element.
If we think that over the timescale of days,
things aren't going to change that much,
we might look towards a strategy of turning on and off the heater.

So dealing with the environmental variability in the niche we're in,
whereas if we're in an open space,
so there's no point in having any strategy of trying to heat the local air
because it's just going to get blown away in the next second,
then you might center decision-making around like finding a warmer location.

And so when there's a more stable environment,
then it puts us into a mode of like understanding the variability patterns,
the circadian patterns and temperature and responding with what we have,
or maybe inventing something new to respond to those regularities.

But when there's the discontinuous and complex environmental settings
that this work is focused on,

it centers this kind of creating one's own niche through action,
not just modifying it more locally to do the best one can.

Cool.

Tyler, do you want to read this claim?

Do you mind?

Sure.

So every organization chooses akin to holding a light torch in the darkness,
which environmental significance acts upon and tend to.

Whatever the torch highlights is equivalent to breaking a quantum wave of probabilities,
moving from superposition to reality,
intermediated by the markup blanket.

In this regard, strategy research is challenged to be less mechanistically determined environment strategy,
but more quantum strategy to environment.

The quantum world is more tenuous than the one imagined.

And this is actually a quote from Ravelli.

This is not Sean.

The quantum world is more tenuous than the one imagined by old physics.

It is made up of happenings.

Get discontinuous events without permanence.

There's a world of fine texture, intricate and fragile as a initial release.

Every interaction is an event.

And it is these light and ephemeral events that weave reality,
not the heavy objects charged with absolute properties,
that are philosophy positive in support of these events.

Cool.

Thanks.

So,

Bishan goes on to say,

and I think we'll come back to this quantum,
because I have some questions here.

So we'll come back to this closer to the end of this section.

But right now,

let's read this claim.

Bishan says,

the critical consideration is how an organism,
or organism,

sorry,

too much biology,

how an organization transmorphs the sensing and sensemaking activity at the individual level into a firm's institutionalized environmental enactment as a basis for continuous strategic renewal. Enactment is essentially an interactive and generative social learning sensemaking process, hence inactive inference.

Dong et al. elegantly address this micro-macro dichotomy.

They say,

while hypothesis setting is an individual activity,

hypothesis testing is an organizational activity.

According to Smircich and Stubbart,

organization members actively form,

enact their environments through social interaction.

A pattern of enactment establishes the foundation of organizational reality,

and in turn has effects in shaping future enactments.

The task of strategic management in this view is organization making,

to create and maintain systems of shared meaning

that facilitate organized action.

Do you guys have any comments here?

One thought,

I'm imagining a fleet of boats.

The upper echelon view,

and the board of directors specifically,

they may not need to be concerned with the minutia of

move the wheel left,

move the wheel right,

on this boat at that time.

But they are,

as it says here,

the task of strategic management in this view is organization making.

So,

that justifies some of this upper echelon focus.

It's not coming,

I don't think,

from a place of elitism,

or chauvinism about those upper management groups,

but it's especially targeting

to those who are in a position to be involved in organization making,

which isn't steering the boat.

But allowing the agency and enactment to happen in the front line requires those not in the front line

to take the view that is being promoted here.

Rather than just imagine if we had enough sensors on enough boats,
we could have all of them viewed up in front of us,
and then we could call out who should turn left and who should turn right.
With the three to eight board meetings per year,
it's not going to happen.

So,

what is the organization making that responds to that look like?

Something I'm kind of curious about what you just said,
because something I've been wondering about through this whole book
is whether a board is really necessary for that meaning making,
or whether that can be done in a more distributed way.

And I think Bajan touched on this a little bit in our last call with him,
saying that,

well,

this is their job,

is to be,

the board's job specifically,

is to be a little bit distinct from the operation.

So,

they're in this unique position to really build meaning and create sensemaking
and facilitate that process.

But I'm wondering if it's actually necessary for that to happen,

because it does strike me as very top down,

imposing a sensemaking framework on an organization is a ton of power.

And it does very much constrain the cognitive abilities of the larger organization.

And it strikes me just on this simple reading of it,

that maybe something is missing from that,

or maybe a potential other way forward that is less top down is possible.

But I'm probably misunderstanding Bajan as well.

So,

I'm curious what y'all think.

It'll be so fun to explore.

But let's just say that the task of strategic management is organization making.

Maybe this is a saddle point view that everybody can get on board with.

But then we go,

great,

upper management's role and function was organization making.

And now everyone has access to the tools and at least view or comment access to the process of organization making.

So,
transmorphing the understanding of what the upper echelon does in terms of organization making.

So,
just double checking,
you're not involved in steering the ship,
you're involved in organization making and action orientation.

And then perhaps that function could be taken on by another organizational structure.
with that as its focus,
or that could be an emergent functionality of other organizational units.

But it's really hard to go from vision coming from the top to simply a vision that's coherent arising from the bottom and from the outside.

And so there's an intermediate view that qualifies the conversation differently.

Yeah,
I think that makes sense.

So,
I think we have observed that sense making happening both in the bottom and the top,
and especially in tech companies,
where like you will have a board and they are making like,
you know,
top down decisions,
but you will also have like consultants almost embedded at each level of the hierarchy who do that sense making at every level of the hierarchy and bubble that up also to the top.

So,
I think it does exist where you kind of have,
it's not purely sense making happening at the board or TMT level.

So,
I just want to point out the first sentence here,
and it really resonates with what you just said,

Tyler,
like the way that I've been reading this.

He says,
the critical consideration is how an organization transmorphs the sensing and sense making activity at the individual level into a firm's institutionalized environmental enactment as a basis for continuous strategic renewal.

Like what I took that like individual level as a very like bottom up thing,
because it's like,
you know,

each like little worker ant,
right,
in the colony.

Like I took that as like a very bottom up and distributed thing.

Like how do you take that and turn it into like some kind of collective sense making,
but really that could also be like,
there's an individual like level at the top,
like each board of directors is selected for their different expertise,
right?

So like you have a variety,
I mean,

ideally you have a variety of different views and perspectives at the board level.

And then how do those individuals like translate their sense making into like something that the institution,
the institution can then act upon.

So I think that that,
that is a very,

like it rings very true that like this individual sensing and sense making is happening at like the top and at the
bottom.

And I think that that's,
that,
that rings true.

So I just want to draw one distinction based on some experience of like doing the top down sense making.
So like I was a strategy consultant for a while.

And a lot of times you would do the top down sense making where you would like talk to all levels of the
hierarchy and you kind of act as the board in that situation and kind of bubble up all of this information into
like a top level idea about like what's happening in the organization,
what the narrative is and what actions they should take.

And so it's like very top down structured,

but there's like other models of doing this where each node in the organization is more or less operating
independently and sense making independently.

Whereas like the top down structure isn't as important.

And maybe you're saying the same thing,

but it strikes me as like a little bit different where sense making isn't happening through a bubble up process
through a pyramid,

but rather happening in like a web,
like in a mesh.

And that strikes me as like an important difference,
but maybe Bajan is saying the same thing.

I'm not sure.

It's hard to know.

And it's hard to also know like how,

you know,

when you have like,

even at like a single layer of the hierarchy,

like you have,

you know,

25 people and like,

you know,

12 of them think one thing and 13 of them think another thing.

Like,

how do you like make sense of that?

Like where there's this like opposing viewpoint,

how do you collectively like get buy-in to proceed with,

like either strategy A or strategy B,

like even at a single level and then pass that information up or down without like,

well,

I wanted it to be this way or I wanted like,

like the people going,

like the opposing view,

like whatever wins,

right?

Like in a vote,

the 13 would win.

And if that's the way that it's run,

or maybe there's weighted voting or who knows,

right?

Like however,

like however a firm would structure itself,

but it's like really important,

I think to get strategy buy-in like from everyone,

at least like hear all of the points in order to proceed and,

um,

have like the viewpoint adopted widely and like get people to get behind it.

So it's,

I don't know,

it's,

it's a hard,

like weird thing to think about how that information is passed up or down in the hierarchy.

All right.

Tyler,

do you have something to add?

Well,

I was just going to add that this is a really important point.

Like I don't think I'm not just trying to be like pedantic about it.

Like it actually,

a lot of organizational strategy debates hinge on this of whether really you can do sense making in a like hierarchical way,

like truly like a pyramid or like a more mesh network.

So like the mesh networks are more like teal organizations or holacracies.

And like,

this is a very contentious point.

And I was hoping that we could like figure it out head on or understand the John's like perspective on it.

And I think he thinks both are possible because I think he's hinted at that in the previous live streams, but he hasn't said it like,

uh,

super clearly so far.

but something I've been like very eagerly waiting for him to like spell out clearly.

So maybe he'll get to it in the next chapters.

So Tyler,

like I have a request,

like in the next live stream,

you have to put in like a slide on holacracy and like explain that to us,

like in a very,

like break it all down,

break down all these different like strategies for us because like you have that expertise and I'm lost.

So that there's,

there's my special request.

Okay.

Awesome.

I mean,

I've just been in like lots of debates with even within,

especially in DAOs where they try to do the mesh network version of things,

not the,

the pyramid.

And a lot of times just like a total disaster and it doesn't work,
but there's always this dream,
especially in DAOs.

Like,

Oh,

could this be possible?

Like,

do you actually need the hierarchy at all?

Can this work more like a spider web or rather than the pyramid?

Um,

and I've been,

this whole book,

I've been like waiting for the answer to that.

So I'm excited to ask for Sean.

Yeah.

And,

and also just getting these dimensions sketched because what if it's like,
well,

we can have a slime mold,

but then we may split a lot and not all the things that split off may survive.

If that is tolerable or preferable to you,
slime mold it is.

But if we all want to ride or die together,
then this kind of a structure is what we want.

So making sure that there's like consent and transparency around what people are wanting to be in,
because also in sort of emergent online 2020s eras, teams and DAOs,
it's like,

are we hanging out in the group chat?

Are we building a product that's going to change the information environment?

Are we learning about other projects?

Are we learning about other early projects to invest in?

Like it can move very rapidly amongst those spaces.

And at the risk of trying to capture four birds and getting none,
there's all these other failure modes,

but maybe again,

that's what those people wanted to do in that blanket at that time.

But when you have shareholders and a board and there are some responsibilities that come into play, and these sort of okay to fail or okay to sunset experiments in online organization, hopefully can reveal empirical patterns that we use to carry forward organizational technique. Yeah, my read so far and what John's doing is that he's saying like, yeah, sure, a slide mold organization is a totally thing you, is a thing you could conceive of.

And like, that's a valid way for thinking about an organization.

But he's specifically thinking about like the legal context that a lot of organizations live in today, and making active inference or free energy governance work for those organizations in the world that we live in.

So I'm not sure if that resonates with you guys, but.

Cool. Okay.

Nice, nice debate.

Tangential, I don't know, information dive.

All right.

So Bijan makes this point and he says organizations are challenged to better distinguish how their actions differ from those of others as they construct eco niches.

And he really kind of gives this like, like a lot of back, a lot of like background information, but I just pulled these two quotes about like distinguishing self from other,

like how do you distinguish what is me and what is you?

And also like about how do you sense make like help others see what's in your head?

And in, in talking about these things, he gives two examples, which I thought were really cool from biology.

So I wanted to read them.

He quotes Godfrey Smith 2016 and says, whenever a fish emits a pulse, it also sends a copy of the command around to the sensing system, enabling that system to counteract the effect of the pulse it has produced.

The fix, the fish is tracking and registering the distinction between self and other between the effects on its senses of its own actions and the effects of events going on around it.

And then he also goes on to say, to use the example of sharks.

He says sharks, he quotes Lanza at out this 2020 paper.

He says sharks have organs called Lorenzini blisters, which sense electrical fields, dolphins and tooth whales, for example, benefit from echolocation abilities.

His sound pulses can penetrate soft tissue to provide them with an X-ray like mental image of the objects of interest in order to tell the other dolphins what they have discovered.

They actually create a visual picture of what they just saw in the minds of other dolphins.

So I didn't like put a full like explanation of echolocation because it gets complicated.

But like, here's this dolphin and it sends out these echolocation rays.

And if something is in its echolocation site, because it's not really site, right?

But if something sees it, then the rays that are reflected back, like give like a, are like molded around.

So you can imagine like ripple ripples through like water, right?

Like, so they mold around like a rock or a stream.

And so what comes back is actually like a physical like ripple picture of the rock or whatever it's seeing.

And so it can actually then modify its own like echolocation sounds to give a picture of the rock to someone else, which I think is so cool.

Right? Like, like to a dolphin friend, not like someone else.

Cause I couldn't see a ripple picture of a rock, but maybe it's dolphin friends could.

Anyway, I think that that is a neat idea of how to like really, you know, we always ask people like, do you understand me?

Do you know what I mean? Do you really know?

But like, it's really hard to know perceptually what someone else is, is what's behind their Markov blanket, right?

Do you guys have anything to add here?

Sorry, I'm just the one fascinated by biological sensing and sense making.

Okay.

Tyler, do you want to read this claim for us?

Sure. So more specifically, optimization is essential across the consistent constituent elements, determining the circular causality underlying inactive inference.

One actions generating reliable sensory input precision to the ability to express beliefs about beliefs consequences.

Sophistication to enact only a bounded set of external states.

Three leveraging salience without missing the very air signaling value that non-salient stimuli may nonetheless behold.

And four, force that to anticipatorily develop and strengthen the internal states, i.e. for resources in form of capabilities critical to resource recombination.

So sensing, sense making and duality management.

Cool.

I find it interesting here that one, two, and three are within the life cycle of an organization or an organism.

They refer to the reliability of action, the ability to have an appropriately expanded vocabulary in terms of sophistication to describe relevant action.

And then three, essentially attention, which is to make sure that you're not just falling victim to hyper skewed or hyper normal stimuli, but rather you're using your attention to enrich incoming stimuli.

And also not ignoring what you're not totally paying attention to that might be on the margins of relevance.

But then four is framed here as an anticipatory question.

In the long run, whether the capabilities are successful to manage resource recombination.

That's the question of life and death.

So one, two, and three are like the behavioral imperatives.

And then four is kind of like the organism looking to the horizon.

Maybe it makes it or maybe its offspring does, or maybe none of the above.

Thank you.

Thank you.

And then just this final quote from the paper or from the book, he says, duality is an essential quantum concept that underpins the mastery of managing the distributed and discontinuous nature of today's market realities.

And like, I'm curious as to like what Bajan is referring to as duality, like in the quantum context, I always think about like wave particle duality.

But I think he has talked here a lot about superposition and maybe he's referring to superposition.

Maybe he means superposition like you can exist in two different states, you know, until the until for a long time, like you can be both black and white one and zero.

So I'm not I'm not really sure what he's referring to as duality, but I think maybe we'll get into it in the next.

Yeah, I was curious whether the connection he's making here is like one of analogies saying like, hey, in the same way, you don't know what was out there in the quantum world.

Do you only you observe something and you kind of collapse all of these waves into an electron?

Is it an analogy in the active inference free energy governance world with through, you know, the inactivist approach?

Or is he saying that this is actually like the same thing that is the same principle underlying?

And I wasn't totally sure which way what way he meant there.

Yes, we should explore. There's the wave particle duality famous.

There's the observer observed duality or the experimenter environments duality.

There's multiple dualities.

And then if we think back to Bayesian physics, what does this help us with?

Let's go into it more in the next chapter and then see how it actually connects back.

Otherwise, I believe it puts us squarely into the quantum business advice.

Genre.

But I know there's more to it.

And that would be like a very sellable like quantum business advice, quantum business strategy manager.

That's my new job.

I'd be like a heck of a consultant title.

Am I working? Am I not working?

Like you could check. You still might not find out.

Am I here? Am I there? Where am I actually?

You can't measure my position and my momentum at the same time.

That's too funny. Okay. All right.

I think we've reached the end of our set of slides.

I'm looking forward to exploring the quantum in more detail as I find the quantum concepts fascinating.

I guess Bayesian also does. Do you guys have anything else to add before we say goodbye?

That's it? No.

Okay. Thanks, Tyler. Thanks, Blue.

We'll see you next week.

Thanks, y'all.

Very well.

Again, we'll see you next week.

Again, we'll see you next week.

Again, we'll see you next week.

Again, we'll see you next week.

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